



Exercise 1

Variables, Collections & Conditionals

Data Analytics with Python · Databricks Environment

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Instructions

Work through each section in your Databricks notebook. Create a new cell for each task. Read each question carefully before writing your code.

- You may refer to the notes and examples shown in class.
- Use `print()` to display your results where asked.
- Write your answers (expected output) in the spaces provided below.
- Complete this document using Microsoft Word or Google Documents.
- Save the document as PDF
- Upload both the PDF and Databricks notebook to GitHub
- Send GitHub link, name, course name to (mvuko@brightlearn.co.za)

The Dataset

Below is a small student dataset. You will manually enter this data as Python variables and collections in your notebook. Each row represents one student's record.

student_name	age	subject	score	passed
"Amara"	19	"Math"	72	True
"Ben"	21	"Science"	55	False
"Chloe"	20	"Math"	88	True
"David"	22	"English"	45	False
"Eva"	19	"Science"	91	True
"Farai"	23	"English"	63	True
"Grace"	20	"Math"	50	False
"Hassan"	21	"Science"	77	True

 **Note:** This is the only data you will work with in this exercise.



Section 1 — Variables & Data Types

In this section, you will create individual variables for a single student's record and explore Python data types.

Task 1.1 — Create variables for Amara's record

In a new cell, create the following variables exactly as shown:

```
student_name = "Amara"  
age          = 19  
subject      = "Math"  
score        = 72  
passed       = True
```

Task 1.1 After running the cell, what is the data type of each variable? Write it next to each variable name.

- a. student_name → _____
- b. age → _____
- c. score → _____
- d. passed → _____

Your answer:

- a. str
- b. int
- c. int
- d. bool

Task 1.2 — Use type() and print()

Add a new cell and use the type() function to confirm the data type of score and passed. Then use print() to display the result.

Task 1.2 Write the code you used and record the output below.

- e. Code: _____
- f. Output for type(score): _____
- g. Output for type(passed): _____

Your answer:

```
print(type(score))  
print(type(passed))  
score : int  
passed 'bool'
```



Task 1.3 — Simple arithmetic

A student's final mark is calculated as their score divided by 100, multiplied by 150 (the exam is out of 150).

Task 1.3 Write a single line of Python that calculates Amara's final mark and stores it in a variable called `final_mark`. Then print it.

- h. What is Amara's `final_mark`? _____
- i. What data type is `final_mark`? _____

Your answer:

```
final_mark = (score / 100) * 150
print(final_mark)
type(final_mark)
final_mark = 108.0
Data type = float
```

Section 2 — Collections

In this section you will store the dataset in Python lists and a dictionary, and access specific values.

Task 2.1 — Create a list of student names

Create a list called `student_names` that contains all 8 student names from the dataset, in order.

Task 2.1 Answer the following questions about your list:

- j. What is `student_names[0]`? _____
- k. What is `student_names[-1]`? _____
- l. What is `student_names[2:5]`? _____
- m. What is `len(student_names)`? _____

Your answer:

```
student_names[0] → "Amara"
student_names[-1] → "Hassan"
student_names[2:5] → ['Chloe', 'David', 'Eva']
len(student_names) → 8
```



Task 2.2 — Create a list of scores

Create a list called scores containing all 8 scores in the same order as the names list.

Task 2.2 Using your scores list, answer:

- n. What is scores[0]? _____
- o. What is scores[4]? _____
- p. What is the result of scores[0] + scores[4]? _____

Your answer:

scores[0] → 72

scores[4] → 91

scores[0] + scores[4] → 163

Task 2.3 — Create a dictionary for one student

Create a dictionary called hassan that stores all five fields for Hassan as key-value pairs.

```
Hassan = {  
    "student_name": ...,  
    "age": ...,  
    "subject": ...,  
    "score": ...,  
    "passed": ...  
}
```

Task 2.3 Using your dictionary, answer:

- q. What does hassan["subject"] return? _____
- r. What does hassan["score"] + 10 return? _____
- s. What does type(hassan["passed"]) return? _____

Your answer:

hassan["subject"] → "Science"

hassan["score"] + 10 → 87

type(hassan["passed"]) → <class 'bool'>



Task 2.4 — A list of dictionaries

Create a list called `students` where each element is a dictionary representing one student. Include only Amara, Ben, Chloe, and David.

Task 2.4 Using your list of dictionaries, answer:

- t. What is `students[1]["score"]`? _____
- u. What is `students[0]["student_name"]`? _____
- v. How many items are in `students`? _____

Your answer:

`students[1]["score"]` → 55

`students[0]["student_name"]` → "Amara"

Number of items → 4

Section 3 — Logical Operators & Conditionals

In this section you will use comparison operators, logical operators (and, or, not), and if/elif/else statements to analyse student records.

Task 3.1 — Comparison expressions

Using the variables you created in Section 1 (Amara's record), evaluate the following expressions. Write the result (True or False) without running the code first — then verify in your notebook.

Task 3.1 Predict the output of each expression:

- w. `score > 60` → _____
- x. `age == 20` → _____
- y. `subject != "Science"` → _____
- z. `score >= 72` → _____

Your answer:

`score > 60` → True

`age == 20` → False

`subject != "Science"` → True

`score >= 72` → True



Task 3.2 — Logical operators

Still using Amara's variables, evaluate these combined expressions:

Task 3.2 Predict and verify:

- aa. `score > 60 and passed == True` → _____
- bb. `age < 18 or score > 70` → _____
- cc. `not passed` → _____
- dd. `score > 80 and subject == "Math"` → _____

Your answer:

- a. True
- b. True
- c. False
- d. False

Task 3.3 — Write an if/elif/else statement

Write a conditional statement that prints a grade category based on a student's score. Use the following rules:

Score Range	Grade
75 and above	Distinction
50 – 74	Pass
Below 50	Fail

Test your conditional using `score = 72` (Amara) and again with `score = 45` (David).

Task 3.3 Write your if/elif/else code below and record the output for each test:

- ee. Output when `score = 72`: _____
- ff. Output when `score = 45`: _____
- gg. Output when `score = 91`: _____

Your answer:

```
def grade(score):  
    if score >= 75:  
        return "Distinction"  
    elif 50 <= score <= 74:  
        return "Pass"  
    else:
```



```
return "Fail"
```

```
print(grade(72))
```

```
print(grade(45))
```

```
print(grade(91))
```

Outputs:

```
score = 72 → "Pass"
```

```
score = 45 → "Fail"
```

```
score = 91 → "Distinction"
```

Task 3.4 — Challenge: combined condition

Write a conditional statement that checks two conditions at once: the student scored above 60 AND their subject is "Math". If both are true, print "Math high achiever". Otherwise print "Does not meet criteria".

Task 3.4 Test with Amara (score=72, subject="Math") and Ben (score=55, subject="Science"):

hh. Output for Amara: _____

ii. Output for Ben: _____

jj. What logical operator did you use? _____

Your answer:

```
def check_math_high(score, subject):
```

```
    if score > 60 and subject == "Math":
```

```
        return "Math high achiever"
```

```
    else:
```

```
        return "Does not meet criteria"
```

```
print(check_math_high(72, "Math"))
```

```
print(check_math_high(55, "Science"))
```

Outputs:

```
Amara → "Math high achiever"
```

```
Ben → "Does not meet criteria"
```

```
Logical operator used → and
```



Reflection

Answer these short questions in your own words — no code needed.

Task R1 What is the difference between a list and a dictionary in Python? Give one example of when you would choose each.

Your answer:

Lists - single variable used to store multiple items. They are ordered, changeable, and allow duplicate values

Dictionaries are used to store data values in key: value pairs. It is a collection which is ordered, changeable and do not allow duplicates

Task R2 In Task 3.2, explain in plain English what this expression checks: `score > 60 and passed == True`

Your answer:

Checks if the student:

Scored above 60 AND

passed the subject

Because we are using an “AND” operator o both conditions must be true.

Task R3 If you wanted to store all 8 students and be able to look up any student by name quickly, which collection type would you use? Why?

Your answer:

Dictionary

Reason: Allows fast lookup by student name (Keys)

Good luck!